

Chapter: SQL Operators & NULL Concepts

The IN Operator

The IN operator checks whether a value belongs to a given set of values. The IN operator checks whether a value belongs to a given set of values.

```
SELECT *  
FROM Students  
WHERE StudentID IN  
(SELECT StudentID  
FROM ScholarshipStudents  
);
```

Meaning

Return every student whose ID appears in the ScholarshipStudents table.

NOT IN & NULL Pitfalls

NOT IN returns rows whose value does not belong to the specified set.

NULL Pitfalls

This is one of the most common SQL interview questions.

Suppose

```
WHERE Salary NOT IN (1000,2000,NULL)
```

Many people expect

All salaries except

- 1000
- 2000

Wrong.

Because SQL evaluates

```
Salary <> 1000
```

AND

```
Salary <> 2000
```

AND

```
Salary <> NULL
```

But `Salary <> NULL` is **UNKNOWN**

The entire condition becomes UNKNOWN, and no rows match

BETWEEN & NOT BETWEEN

BETWEEN checks whether a value lies within a specified range.

```
SELECT *  
FROM Students  
WHERE Name BETWEEN 'A' AND 'M';
```

NOT BETWEEN

Example

```
SELECT *  
FROM Employees  
WHERE Salary NOT BETWEEN 50000 AND 100000;
```

What is NULL?

Definition

NULL meansIt means:

The database does not know the value, or the value does not exist.

Three-Valued Logic

SQL has three logical states.

Result	Meaning
TRUE	Condition satisfied
FALSE	Condition not satisfied
UNKNOWN	Cannot determine because of NULL

Three-Valued Logic with OR

A	B	A OR B
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
TRUE	UNKNOWN	TRUE
FALSE	TRUE	TRUE
FALSE	FALSE	FALSE
FALSE	UNKNOWN	UNKNOWN
UNKNOWN	TRUE	TRUE
UNKNOWN	FALSE	UNKNOWN
UNKNOWN	UNKNOWN	UNKNOWN

Aggregate Functions with NULL

Consider the following Salaries:

Salary
1000
2000
NULL

COUNT(*)

```
SELECT COUNT(*)
```

Result 3

Counts rows.

COUNT(Salary)

```
SELECT COUNT(Salary)
```

Result 2

Ignores NULL.

AVG

```
SELECT AVG(Salary)
```

Average

$(1000+2000)/2 = 1500$

NULL is ignored.

SUM

```
SUM(Salary)
```

Result 3000

NULL is ignored.

MIN/MAX

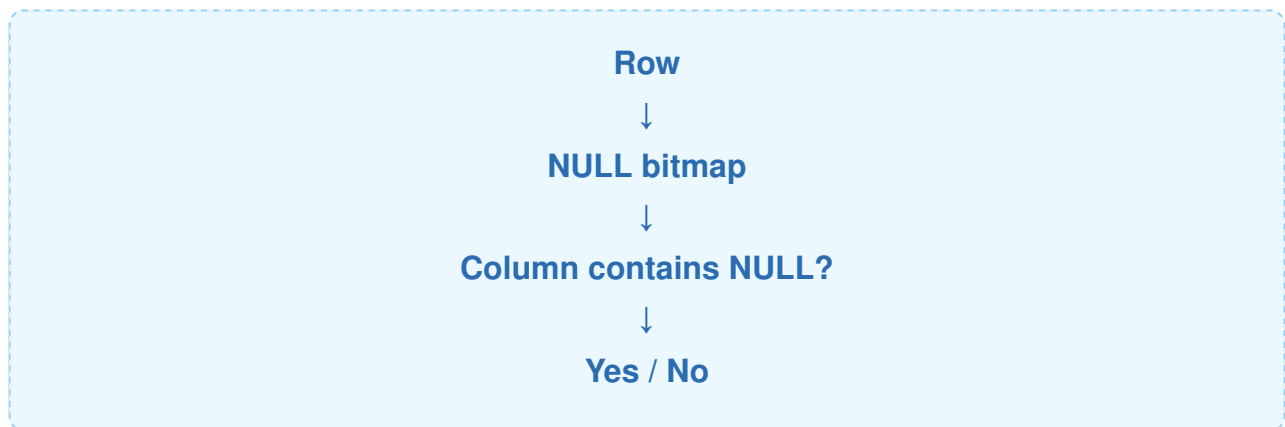
NULL values are ignored.

Internal Representation & Performance

Internal Representation

Internally, MySQL stores a NULL bitmap for nullable columns.

Conceptually, think of it as:



This allows the database to know whether a column contains a real value or not.

Performance Considerations

Searching with

```
WHERE Column IS NULL
```

can use an index if one exists on the column (COUNT(*) counts all rows.

COUNT(column) ignores NULL values.